Four roots of Yuster's entropy polynomial at $k = 50$

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Abstract

An exact rational sign certificate disproves Conjecture 3.9 of Yuster: p_{50} has at least four distinct roots in $(0, 1)$. The certificate uses integer arithmetic and a standard multisection formula, also appearing in Wakhare's work. It concerns the polynomial root conjecture; the associated entropy inequality has already been proved by Ho.

Task 1: complete disproof

For an integer $k \geq 2$, let $\varphi_k \in (0, 1)$ satisfy $\varphi_k^k + \varphi_k = 1$, and put $\alpha_k = \varphi_k^{k-1} = \varphi_k^{-1} - 1$. Equivalently, α_k is the unique positive solution of

$$\alpha_k(1 + \alpha_k)^{k-1} = 1. \tag{1}$$

Set $h(x) = -x \log x - (1 - x) \log(1 - x)$ and $f_k(x) = \alpha_k h(x^k) - x^{k-1} h(x)$ on $(0, 1)$. Yuster's polynomial is characterized by his Corollary 3.7 [1]:

$$f_k^{(k+1)}(x) = \frac{(k-1)! p_k(x)}{x(x^k - 1)^k}. \tag{2}$$

This is exactly the normalization in the original conjecture.

Theorem 1. *The polynomial p_{50} has a root in each of*

$$(0, 1/4), \qquad (1/4, 1/2), \qquad (1/2, 3/5), \qquad (3/5, 1).$$

In particular, the assertion that every p_k has at most two roots in $(0, 1)$, counting multiplicity, is false.

A convenient exact polynomial formula

Define

$$S_k(x) = \sum_{i=0}^{k-1} x^i, \quad B_k(x) = \sum_{i=0}^{k-1} (1 - x)^i, \quad H_k(y) = \sum_{j=0}^{k-1} a_{k,j} y^j,$$

where

$$a_{k,j} = [z^{kj}] S_k(z)^k = \sum_{v=0}^{j} (-1)^v \binom{k}{v} \binom{k(j - v) + k - 1}{k - 1}. \tag{3}$$

Every $a_{k,j}$ is positive: kj can be written as a sum of k integers between 0 and $k - 1$. In particular $H_k(y) > 0$ for $y \geq 0$. These coefficients and the following derivative identities are known; see Wakhare's Corollaries 13 and 14 [2]. We include the derivation to fix all signs and normalizing factors.

Lemma 1. *One has the polynomial identity*

$$p_k(x) = (-1)^k\big(B_k(x)S_k(x)^k - \alpha_k k^2 H_k(x^k)\big). \tag{4}$$

Proof. Taking coefficients at multiples of k in $(1-z)^{-k} = S_k(z)^k/(1-z^k)^k$ gives

$$\sum_{n\ge 0}\binom{kn+k-1}{k-1}x^{kn} = \frac{H_k(x^k)}{(1-x^k)^k}. \tag{5}$$

The convergent expansion $h(u) = -u\log u + u - \sum_{n\ge 1} u^{n+1}/(n(n+1))$ may be differentiated term by term on compact subintervals of $(0,1)$. It yields

$$\frac{d^{k+1}}{dx^{k+1}}h(x^k) = -\frac{k\,k!}{x}\sum_{n\ge 0}\binom{kn+k-1}{k-1}x^{kn},$$
$$\frac{d^{k+1}}{dx^{k+1}}\big(x^{k-1}h(x)\big) = \frac{(k-1)!}{x^2}\big(1-(1-x)^{-k}\big).$$

Combining these equations with (5), using $B_k(x) = (1-(1-x)^k)/x$ and $S_k(x) = (1-x^k)/(1-x)$ on $(0,1)$, gives

$$f_k^{(k+1)}(x) = \frac{(k-1)!}{x(1-x^k)^k}\big(B_k(x)S_k(x)^k - \alpha_k k^2 H_k(x^k)\big).$$

Comparison with (2) proves (4) on $(0,1)$, hence identically as a polynomial. □

Exact signs

For $0 \le x \le 1$, put

$$R_k(x) = \frac{B_k(x)S_k(x)^k}{k^2 H_k(x^k)}.$$

The denominator is positive. For even k, the sign of $p_k(x)$ is the sign of $R_k(x) - \alpha_k$. Since $t(1+t)^{49}$ is strictly increasing for $t > 0$, the integer inequalities

$$59\cdot 1059^{49} < 1000^{50}, \qquad 3\cdot 53^{49} > 50^{50}$$

give

$$\frac{59}{1000} < \alpha_{50} < \frac{3}{50}. \tag{6}$$

The following rational bounds follow directly from (3):

x	Exact value or strict rational enclosure of $R_{50}(x)$	$\operatorname{sgn} p_{50}(x)$
0	$1/50$	$-$
$1/4$	$(2717, 2718)$	$+$
$1/2$	$(1/50, 21/1000)$	$-$
$3/5$	$(31/500, 63/1000)$	$+$
1	$1/50$	$-$

For completeness, the entire arithmetic certificate for this table is reproducible by the following Python code, using exact fractions only. Thus none of the strict signs depends on a numerical root finder.

```
from fractions import Fraction as Q
from math import comb
k = 50
a = [sum((-1)**v*comb(k,v)*comb(k*(j-v)+k-1,k-1)
         for v in range(j+1)) for j in range(k)]
def R(x):
    S = sum(x**i for i in range(k))
    B = sum((1-x)**i for i in range(k))
    H = sum(Q(c)*x**(k*j) for j,c in enumerate(a))
    return B*S**k/(k*k*H)
assert 59*1059**49 < 1000**50
assert 3*53**49 > 50**50
assert R(Q(0)) == R(Q(1)) == Q(1,50)
assert 2717 < R(Q(1,4)) < 2718
assert Q(1,50) < R(Q(1,2)) < Q(21,1000)
assert Q(31,500) < R(Q(3,5)) < Q(63,1000)
```

The accompanying `exact-certificate.json` stores the coefficients and exact fractions. The separate `verify.py` additionally reconstructs the coefficients by bounded-composition convolution and checks the original C recurrence and ρ_k, σ_k sums from [1] at all five points, including $k = 50$.

Proof of Theorem 1. The table and (6) give the strict sign sequence $-, +, -, +, -$. The intermediate value theorem, applied on each of the four adjacent closed intervals, gives a root in its interior. The four open intervals are disjoint. These are therefore four distinct roots of the polynomial specified by (2), proving the assertion. □

Scope and prior results

Yuster proved the root claim for $k = 2, 3, 4$ and used it as a sufficient condition for an entropy inequality. Ho [3] later proved that entropy inequality for every real $k > 1$ by a different argument. The present counterexample concerns the sufficient root condition. The derivative and multisection identities used above are explicitly credited to the prior literature. No claim is made here that 50 is the least counterexample or that p_{50} has exactly four roots. Wakhare's polynomial with parameters $(k, r) = (50, 1)$ differs from p_{50} by the nonzero factor $-(1-x)^{50}/50$ on $(0, 1)$; thus the same witness also refutes his broader Conjecture 2. This consequence receives no separate campaign credit.

Tasks 2–5: proposed follow-up work

2. Determine the least integer $k \geq 2$ for which p_k has more than two roots in $(0, 1)$, with an exact certificate for every smaller k.

3. Prove that the number of distinct roots of p_k in $(0, 1)$ is unbounded as $k \to \infty$.

4. Find asymptotic locations of the successive root pairs, using the multisection sum in (5).

5. Classify the coprime integer pairs $k > r \geq 1$ for which Wakhare's entropy polynomial $p_{k,r}$ has exactly two roots in $(0, 1)$.

References

[1] R. Yuster, *Almost k-union closed set systems*, arXiv:2302.12276v1 (2023), especially Corollary 3.7 and Conjecture 3.9. `https://arxiv.org/abs/2302.12276`.

[2] T. Wakhare, *Iterated entropy derivatives and binary entropy inequalities*, Journal of Approximation Theory **307** (2025), 106143. Complete arXiv version read: `https://arxiv.org/abs/2312.14743`, v2, Corollaries 13–14.

[3] B. S. Ho, *A generalization of Boppana's entropy inequality*, arXiv:2601.19327v1 (2026). `https://arxiv.org/abs/2601.19327`.